MSG No. 88.82 $I4_1/a1'$ [Type II, tetragonal]

Table 1: Wyckoff site: `4a`, site symmetry: `-4..1'`

No.	position	mapping
1	$\left[0, \frac{1}{4}, \frac{1}{8}\right]$	`[1,4,14,15,17,20,30,31]`
2	$\left[\frac{1}{2}, \frac{1}{4}, \frac{3}{8}\right]$	`[2,3,13,16,18,19,29,32]`
3	$\left[0, \frac{3}{4}, \frac{7}{8}\right]$	`[5,8,10,11,21,24,26,27]`
4	$\left[\frac{1}{2}, \frac{3}{4}, \frac{5}{8}\right]$	`[6,7,9,12,22,23,25,28]`

Table 2: Wyckoff site: `4b`, site symmetry: `-4..1'`

No.	position	mapping
1	$\left[0, \frac{1}{4}, \frac{5}{8}\right]$	`[1,4,14,15,17,20,30,31]`
2	$\left[\frac{1}{2}, \frac{1}{4}, \frac{7}{8}\right]$	`[2,3,13,16,18,19,29,32]`
3	$\left[0, \frac{3}{4}, \frac{3}{8}\right]$	`[5,8,10,11,21,24,26,27]`
4	$\left[\frac{1}{2}, \frac{3}{4}, \frac{1}{8}\right]$	`[6,7,9,12,22,23,25,28]`

Table 3: Wyckoff site: `8c`, site symmetry: `-11'`

No.	position	mapping
1	$[0, 0, 0]$	`[1,5,17,21]`
2	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}\right]$	`[2,14,18,30]`
3	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$	`[3,15,19,31]`
4	$\left[0, \frac{1}{2}, 0\right]$	`[4,8,20,24]`
5	$\left[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}\right]$	`[6,10,22,26]`
6	$\left[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}\right]$	`[7,11,23,27]`
7	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[9,13,25,29]`
8	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[12,16,28,32]`

Table 4: Wyckoff site: `8d`, site symmetry: `-11'`

No.	position	mapping
1	$\left[0, 0, \frac{1}{2}\right]$	`[1,5,17,21]`
2	$\left[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}\right]$	`[2,14,18,30]`
3	$\left[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}\right]$	`[3,15,19,31]`
4	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[4,8,20,24]`
5	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}\right]$	`[6,10,22,26]`
6	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}\right]$	`[7,11,23,27]`
7	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[9,13,25,29]`
8	$\left[\frac{1}{2}, 0, 0\right]$	`[12,16,28,32]`

Table 5: Wyckoff site: `8e`, site symmetry: `2..1'`

No.	position	mapping
1	$\left[0,\ \frac{1}{4},\ z\right]$	`[1,4,17,20]`
2	$\left[\frac{1}{2},\ \frac{1}{4},\ z+\frac{1}{4}\right]$	`[2,3,18,19]`
3	$\left[0,\ \frac{3}{4},\ -z\right]$	`[5,8,21,24]`
4	$\left[\frac{1}{2},\ \frac{3}{4},\ \frac{3}{4}-z\right]$	`[6,7,22,23]`
5	$\left[\frac{1}{2},\ \frac{3}{4},\ z+\frac{1}{2}\right]$	`[9,12,25,28]`
6	$\left[0,\ \frac{3}{4},\ z+\frac{3}{4}\right]$	`[10,11,26,27]`
7	$\left[\frac{1}{2},\ \frac{1}{4},\ \frac{1}{2}-z\right]$	`[13,16,29,32]`
8	$\left[0,\ \frac{1}{4},\ \frac{1}{4}-z\right]$	`[14,15,30,31]`

Table 6: Wyckoff site: `16f`, site symmetry: `11'`

No.	position	mapping
1	$[x,\ y,\ z]$	`[1,17]`
2	$\left[\frac{3}{4}-y,\ x+\frac{1}{4},\ z+\frac{1}{4}\right]$	`[2,18]`
3	$\left[y+\frac{1}{4},\ \frac{1}{4}-x,\ z+\frac{1}{4}\right]$	`[3,19]`
4	$\left[-x,\ \frac{1}{2}-y,\ z\right]$	`[4,20]`
5	$[-x,\ -y,\ -z]$	`[5,21]`
6	$\left[y+\frac{1}{4},\ \frac{3}{4}-x,\ \frac{3}{4}-z\right]$	`[6,22]`
7	$\left[\frac{3}{4}-y,\ x+\frac{3}{4},\ \frac{3}{4}-z\right]$	`[7,23]`
8	$\left[x,\ y+\frac{1}{2},\ -z\right]$	`[8,24]`
9	$\left[x+\frac{1}{2},\ y+\frac{1}{2},\ z+\frac{1}{2}\right]$	`[9,25]`
10	$\left[\frac{1}{4}-y,\ x+\frac{3}{4},\ z+\frac{3}{4}\right]$	`[10,26]`
11	$\left[y+\frac{3}{4},\ \frac{3}{4}-x,\ z+\frac{3}{4}\right]$	`[11,27]`
12	$\left[\frac{1}{2}-x,\ -y,\ z+\frac{1}{2}\right]$	`[12,28]`
13	$\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ \frac{1}{2}-z\right]$	`[13,29]`
14	$\left[y+\frac{3}{4},\ \frac{1}{4}-x,\ \frac{1}{4}-z\right]$	`[14,30]`
15	$\left[\frac{1}{4}-y,\ x+\frac{1}{4},\ \frac{1}{4}-z\right]$	`[15,31]`
16	$\left[x+\frac{1}{2},\ y,\ \frac{1}{2}-z\right]$	`[16,32]`